Palaeogeography, Palaeoclimatology, Palaeocology
Elsevier Publishing Company, Amsterdam – Printed in The Netherlands

THE ROTATION OF SPAIN: PALAEOMAGNETIC EVIDENCE FROM THE SPANISH MESETA

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(Received February 2, 1967)

SUMMARY

The magnetic properties of 39 oriented samples of Triassic siltstones and of 43 oriented samples of Silurian andesites and basalts from the Spanish Meseta were investigated.

After a.c. magnetic field demagnetization up to 3,000 Oe, peak value, and after correction for the dip of the geological strata the Triassic samples revealed a characteristic direction of magnetization with an average declination (*D*) of 359.5° and an inclination (*I*) of 23°. One group of samples of Silurian volcanic rocks from the province of Guadalajara had, after a.c. magnetic field demagnetization and correction for the dip, a characteristic direction of magnetization with an average declination of 159° and an inclination of 18.5°; another group from the province of Ciudad Real had, after a.c. magnetic field demagnetization and after correction for the dip, a characteristic direction of magnetization with an average declination of 130.5° and an inclination of 22.5°.

The palaeomagnetic pole position derived from the Triassic rocks is situated at 63°N 177.5°E, and the pole positions of the first and second group of Silurian samples are situated at 35.5°N 157°W and 21°N 132°W, respectively.

The divergency between these results and the already known data for Triassic and Silurian from stable (extra-Alpine) Europe highly supports a counterclockwise rotation of Spain.

INTRODUCTION

The present study forms part of the palaeomagnetic investigations carried out by members of the State University, Utrecht, The Netherlands, under direction of Professors M. G. Rutten, R. W. van Bemmelen and J. Veldkamp. Since 1960 a special topic of these investigations has been the palaeomagnetic analysis of rocks from the Alpine belts of Europe (DIETZEL, 1960; VAN DER LINGEN, 1960; VAN

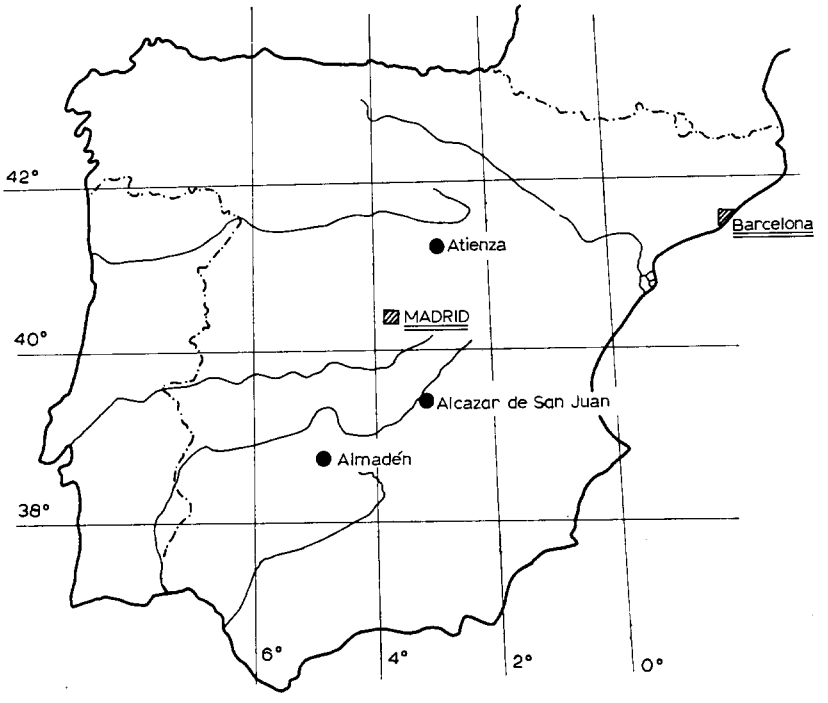
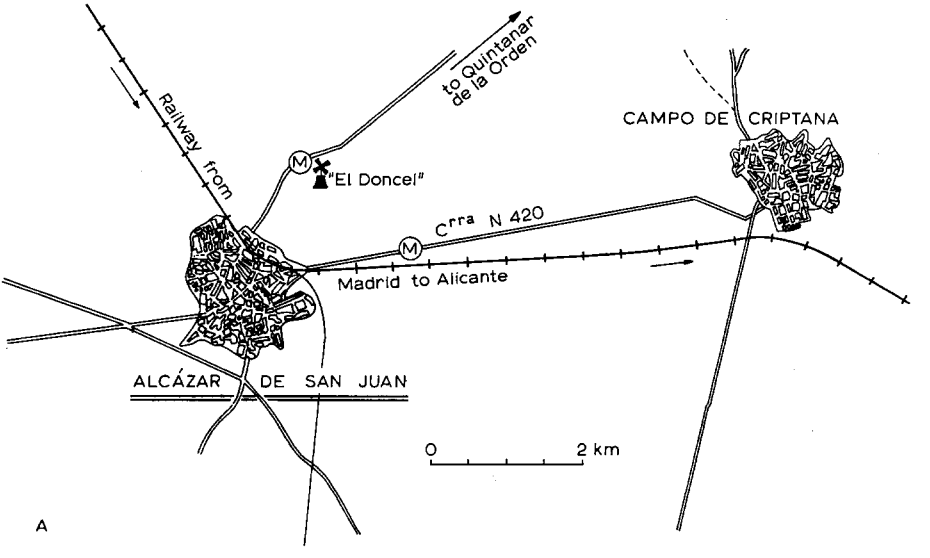


Fig.1. Regional map of the Iberian peninsula showing sample areas of Fig.2.



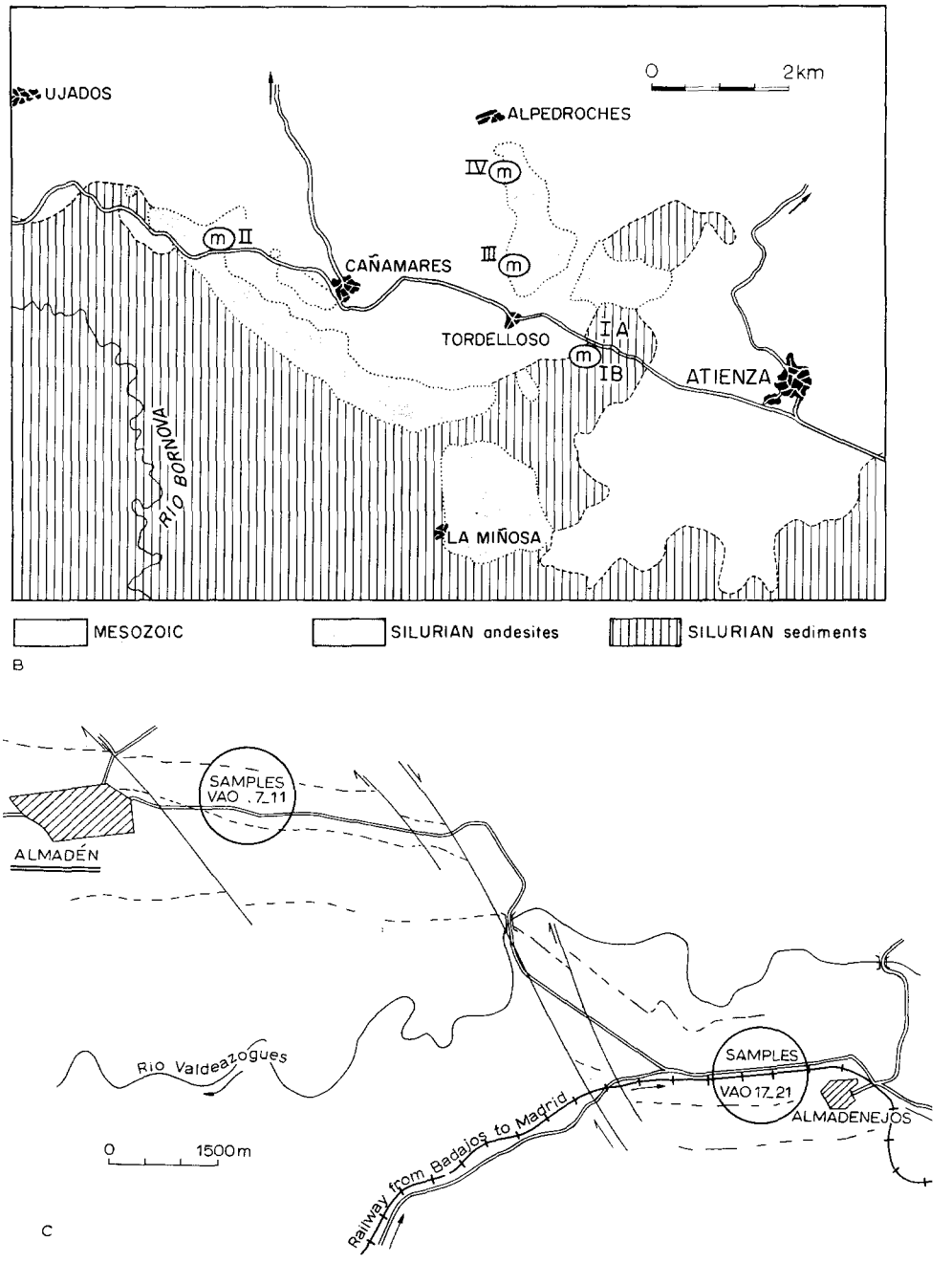


Fig.2. Maps of the Spanish sample areas. A. Alcázar de San Juan (Triassic); samples VAL 1–34, ALC 1–3, ALC 21–23; B. Atienza (Silurian); samples VAN 1–30, CAN 1–23; C. Almadén (Silurian); samples VAO 7–11, VAO 17–21.

HILTEN, 1960, 1962; SCHWARZ, 1962, 1963; DE BOER, 1963, and GUICHERIT, 1964). In particular Schwarz and Van der Lingen found palaeomagnetic indications from Permo-Triassic rocks in the central Pyrenees that affirmed an hypothesis of various papers on geotectonic processes: a counter-clockwise rotation of 30–40° of the Iberian block (CAREY, 1958; BULLARD et al., 1965; GIRDLER, 1965).

Recent investigations (VAN DONGEN, 1967; and the present paper) on volcanic rocks and sediments in Spain can be compared with the results mentioned above from the central Pyrenees.

In the summer of 1963 and 1966 about 200 oriented samples of Triassic siltstones and sandstones were collected from the redbed formations on the eastern margin of the Spanish Meseta. The results of 39 samples collected near Alcázar de San Juan, 140 km south of Madrid, are described in this paper; the results of the other samples, which contained almost only present-day geomagnetic field directions and had very low intensities of N.R.M., will be published later (VAN DER VOO, in preparation). Moreover near Atienza, 140 km north of Madrid, 33 samples were collected from four flows (six sites) of Upper Silurian andesites and for comparison with the results obtained from these rocks, the region of Almadén, 210 km southwest of Madrid, was visited and ten samples from two sites of Upper Silurian volcanic rocks were taken.

The three regions visited are indicated in Fig.1 and the sample sites are given on the maps of Fig.2.

GEOLOGY OF THE REGIONS INVESTIGATED

The Triassic of Alcázar de San Juan

Situated on the eastern margin of the Spanish Meseta, the Palaeozoic basement with its Triassic sedimentary cover forms here an outcrop in the formations of the recent intramontane basins. The flatlying basal redbeds, mainly consisting of marls, siltstones and coarse-grained sandstones are considered to be of Lower Triassic age (KINDELÁN Y DUANY, 1952).

Stratigraphically their sedimentation was followed by the deposition of Muschelkalk (Middle Triassic) and Keuper marls (Upper Triassic), both locally fossiliferous. Samples have been collected from two sites in different parts of the stratigraphical column of the basal redbeds, each site covering about 3 m sediment.

The Silurian of Atienza

The area has been investigated by SCHRÖDER (1930), who geologically mapped the Hercynian-folded Palaeozoic formations and its Mesozoic cover of sediments from Triassic, Jurassic and Cretaceous age, gently folded (Alpine) into synclines and anticlines with a southwest–northeast trend (Guadarrama trend)

in the south and an east-southeast–west-northwest trend (Hesperian chains) in the north.

The Silurian effusive rocks of andesitic type are found at the extreme northern boundary of the Silurian outcrops, which form a dome structure consisting of quartzites and shales around a nucleus of gneisses near Hiendelaencina, approximately 15 km south of Atienza. With some variation the andesites have a moderate dip towards the north.

Samples have been collected from at least four different flows (see Table I, flows *I*, *II*, *III* and *IV*).

The Silurian of Almadén

Due to the occurrence of mineral veins the region has been studied by several geologists (CALDERÓN Y ARANA, 1884; VAN DER VEEN, 1924; ALMELA, 1959; ALMELA and FEBREL, 1960). The Hercynian trend of isoclinal folding and faulting is west-northwest–east-southeast, and the mainly Silurian and Devonian sediments with intercalated volcanic rocks have a subvertical dip. The first sampling site is situated in the immediate neighbourhood of Almadén, where five samples have been collected from one flow of the basaltic rocks intercalated between Silurian quartzites and fossiliferous shales. Five more samples have been taken from one dacitic flow near the railway station of Almadenejos, situated at 11.5 km east of Almadén.

METHODS OF RESEARCH

The samples were sawn to approximately equidimensional shape and embedded in their correct orientation in cubes of paraffin with 10–cm edges; thereupon directions and intensities were measured on the Utrecht astatic magnetometers. The natural remanent magnetization (N.R.M.) of all samples was analysed by progressive demagnetization with a.c. magnetic fields up to 3,000 Oe (peak value). For an extensive description of the methods of measurement and the analysis of the N.R.M. with the aid of a.c. magnetic field demagnetization the reader is referred to As and ZIJDERVELD (1958), As (1960), and ZIJDERVELD (1967a).

The numerical data thus obtained are summarized in Table I and II, and Fig.3, 5 and 7 show the data plotted in stereographic projection. The correction for the geomagnetic variation has been applied.

RESULTS OF THE TRIASSIC SAMPLES (SAMPLES VAL 1–34, ALC 1–3, ALC 21–23)

The intensity of the N.R.M. of the siltstones varied between 21.10^{-6} and 4.10^{-6} e.m.u./cm^3.

TABLE I

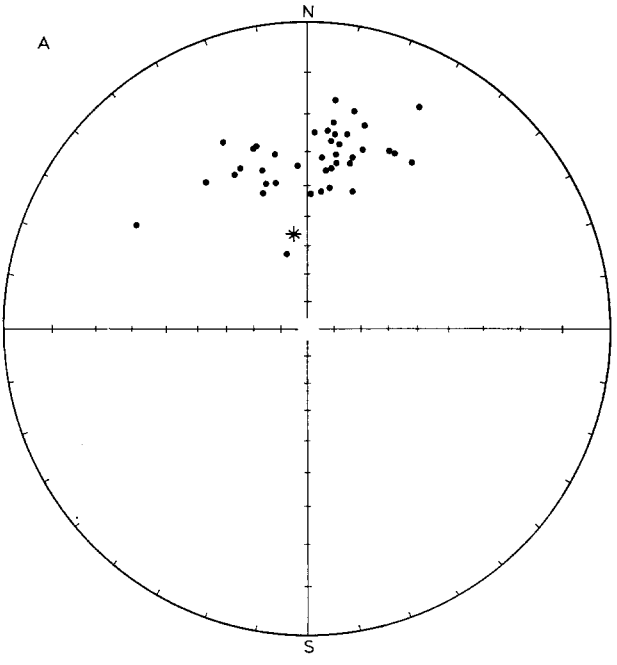
SUMMARY OF MEASUREMENTS: NUMERICAL DATA FROM THE SAMPLES

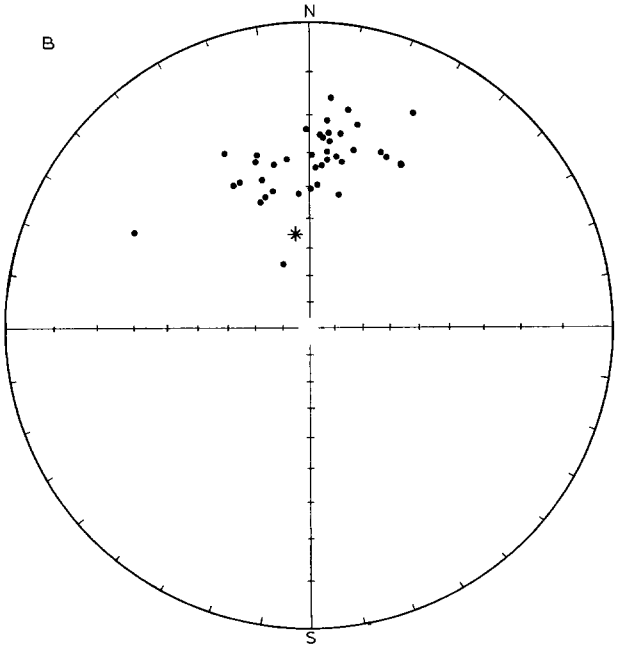
Site	*Sample No.*	*N.R.M.*[1]				*Characteristic component*[1]		*Characteristic component after correction for the dip*[1]	
		D	*I*	*J/cm*[3]	*Q-value*	*D*	*I*	*D*	*I*
1 El Doncel	ALC 21	5.5	24	12.60	3.30	2	21.5	359.5	21
	ALC 22	1.5	42.5	7.99	5.32	356	36	353.5	35.5
	ALC 23	18	40	12.83	8.02	13	34	9.5	33.5
	VAL 1	9	34.5	10.45	4.83	5	29	1	28.5
	VAL 2	27	12	8.88	28.31	20	10	18	11.5
	VAL 3	357	33	9.31	3.51	357.5	30.5	353.5	29
	VAL 4	7	16	10.65	3.33	6.5	13.5	5	13
	VAL 5	12.5	18	9.88	4.61	6	11	4.5	10.5
	VAL 6	12	24.5	14.61	3.90	10.5	22	8	22
	VAL 7	25	25	8.44	4.45	23	21.5	20	22.5
	VAL 8	9.5	27.5	11.52	4.84	5.5	15.5	3.5	15
	VAL 9	7.5	26.5	9.80	3.68	—	—	—	—
	VAL 10	32.5	24.5	7.83	4.27	22.5	16	20.5	18
	VAL 11	16	21	8.69	2.81	13	14.5	11	15
	VAL 12	4.5	32	7.60	3.91	4	13	2.5	12.5
	VAL 13	14	30.5	10.91	3.26	8.5	24.5	5.5	24.5
	VAL 14A	27	25	19.54	7.00	18	20	15	21.5
	VAL 14B	17.5	27.5	20.67	9.60	10	22.5	6.5	22
	VAL 15	10	32.5	13.60	3.95	9	28.5	5.5	28.5
	VAL 16	7	21	11.86	3.33	3.5	17.5	1.5	17
	VAL 17	8	25	21.22	8.82	4.5	21	2	20
	VAL 18	2	24.5	10.63	3.24	1.5	17.5	359	16
	VAL 19	15	30	20.77	5.62	16	30.5	11.5	30.5
	VAL 22	7	34.5	8.08	3.31	1.5	27	358	25.5
	VAL 23	5.5	41.5	4.45	4.66	9	40.5	3	39.5
	VAL 24	9	40.5	6.03	3.43	9	36	4	35
	VAL 25	9.5	30.5	7.52	2.92	9.5	26	6	26
2 Crra N 420	ALC 1	336	31	5.54	6.90	348	14.5	347.5	17
	ALC 2	343	38	4.89	6.51	2.5	23	2	25.5
	ALC 3	343	27	7.14	4.31	1	− 3.5	1	0
	VAL 26	335	22.5	4.15	2.51	354.5	15	354.5	18
	VAL 27	334	32	7.16	3.39	353	17.5	353	20
	VAL 28	344	26.5	7.88	3.06	354.5	16.5	355	19
	VAL 29	344	33	7.51	4.67	352.5	13	352.5	15
	VAL 30	349	30	8.13	3.71	352.5	18	351.5	21
	VAL 31	341	40	7.69	5.22	354	31	352	33
	VAL 32	344	62	4.95	3.30	354.5	36	353	38
	VAL 33	300.5	23.5	5.48	5.87	343	21.5	342	24
	VAL 34	347	37.5	5.03	2.89	357.5	34	356	36
I A Tordelloso	CAN 1	88	11	74.8	1.15	107.5	−19.5	108	10
	CAN 2	133	− 7	300.2	2.34	134	−12	134	18
	CAN 3	139	−13.5	180.0	2.05	136	−15.5	136	14
	CAN 4	127.5	− 3	99.5	2.38	131	−10	133	16.5
	CAN 6A	341	51.5	67.6	3.86	137	− 9.5	137.5	20
	CAN 6B	327	49.5	112.1	4.02	133	−13	133	16.5
	CAN 7	142	12	312.8	1.42	137.5	− 6	138.5	23.5

TABLE I (*continued*)

Site	*Sample No.*	*N.R.M.*[1]				*Characteristic component*[1]		*Characteristic component after correction for the dip*[1]	
		D	*I*	*J/cm*³	*Q-value*	*D*	*I*	*D*	*I*
I B	CAN 8	208	5.5	128.2	0.85	176.5	−10.5	176.5	18
Tordelloso	CAN 9	34.5	76.5	68.1	0.44	160	2	159	30
	CAN 12	173.5	−19.5	140.6	4.59	179	−20.5	178.5	8.5
	CAN 13	181	− 9	272.2	3.83	179	−20	178.5	9.5
II	CAN 14	161.5	− 6	461.8	10.91	162	− 8	160.5	12
Peña Negra	CAN 16	153.5	− 2.5	7,708.0	13.28	155.5	− 4	153.5	11
	CAN 18	166	− 9.5	177.4	6.59	167	−11	166.5	14
	CAN 21	160.5	−11.5	2,321.1	12.69	160.5	−15	164.5	6
	CAN 23	162	− 7	100.7	9.09	161	− 6	158.5	13.5
III	VAN 3	161	2.5	63.7	10.40	161	1	158	26
Tordelloso	VAN 4	161	−14	69.4	9.05	161	−13.5	161.5	13.5
	VAN 5	164	− 1	80.8	9.72	163	− 8	162.5	19.5
	VAN 6	164	− 7	80.4	11.91	162	− 9	161	19
	VAN 7	161.5	− 5,5	69.1	9.06	161.5	− 6	159.5	21
	VAN 8	167	− 4.5	72.6	9.96	167	− 4.5	165	24
	VAN 10	165.5	− 6.5	70.7	8.90	161.5	− 7	161	20
IV	VAN 11	156	8	293.5	10.37	155.5	5.5	150.5	30
Tordelloso	VAN 12	173	−13	359.5	12.31	170.5	− 6	169	23
	VAN 14	175	0	267.8	11.24	174	− 3.5	172	26
	VAN 17	166	7	337.2	11.38	166	2.5	162.5	30
	VAN 19	163.5	5	318.9	8.38	166.5	1	163.5	28.5
II	VAN 24	170	−13.5	3,129.0	17.46	174	−11	172.5	18
Peña Negra	VAN 26	162.5	−10	98.6	9.23	162.5	−10.5	162.5	11
	VAN 28	162.5	− 5	88.6	8.09	162.5	− 5	158.5	15.5
	VAN 29	166.5	− 7.5	88.1	8.07	165.5	− 6	162	14
	VAN 30	167.5	− 5	97.4	9.34	166	− 5.5	161.5	18
V	VAO 7	51	46	558.6	0.46	100	18.5	119	8
Almadén	VAO 8	48.5	44	591.3	0.75	74.5	24	119	32.5
	VAO 9	54	52.5	583.9	0.70	91.5	28	126	18
	VAO 10	54.5	52	568.6	0.61	90	32.5	130.5	20.5
	VAO 11	41	36	431.4	0.52	85.5	13.5	108	20
VI	VAO 17	7.5	41	4.08	0.17	26	31.5	137	20
Almade-	VAO 18	355	53	1.93	0.10	354.5	43.5	165	17
nejos	VAO 19	8.5	16.5	30.20	0.72	17.5	1	122.5	49
	VAO 20	26	44.5	1.35	0.09	29.5	36.5	137	14.5
	VAO 21	16	51	1.15	0.06	27	23.5	131.5	25

[1] Corrected for the present-day local geomagnetic declination, D = declination in degrees; I = inclination in degrees; J/cm^3 = intensity in 10^{-6} e.m.u./cm³; Q-value = $J/k.H$, (k = susceptibility; H = the strength of the ambient field; in the case of the measuring method $H = H_z$, the vertical component of the ambient field).


A
N
S


B
N
S

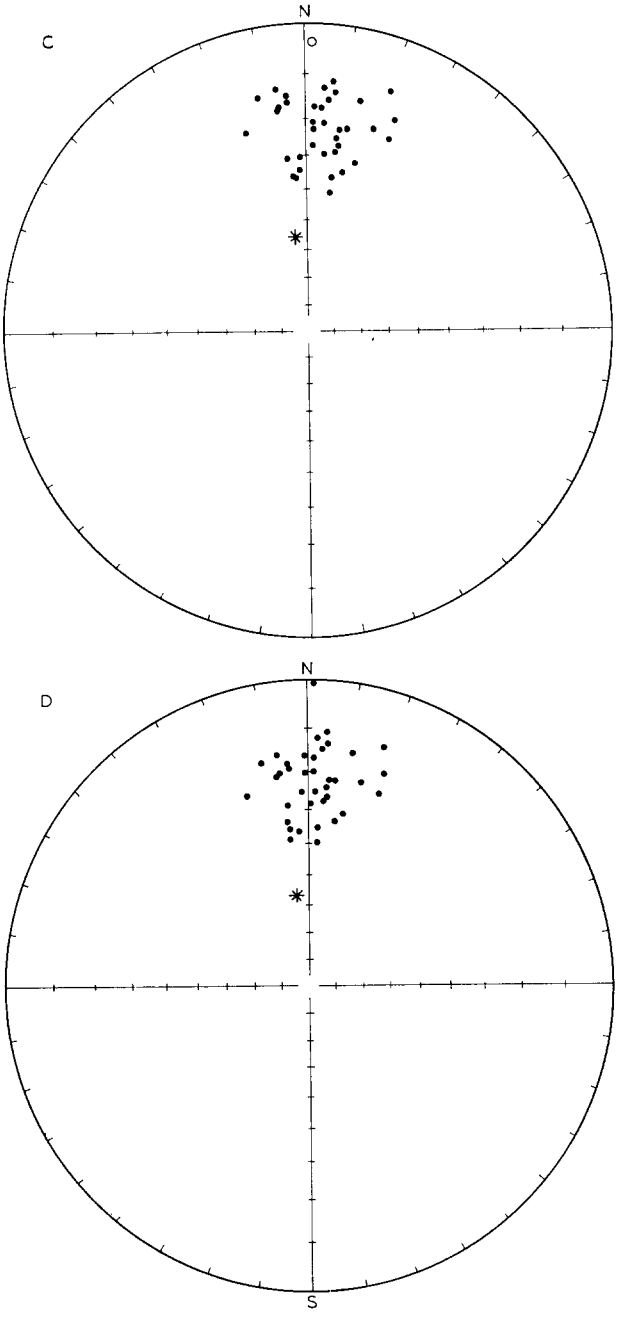


Fig.3. Triassic siltstones from Alcázar de San Juan. Stereograms showing the directions of magnetic vectors: A. Natural remanent magnetization, without tectonic correction; B. Natural remanent magnetization, after tectonic correction; C. Components, removed between 1,000 and 3,000 Oe, without tectonic correction; D. Components, removed between 1,000 and 3,000 Oe, after tectonic correction.

● = north-seeking poles pointing down; ○ = north-seeking poles pointing upwards. ★ = the direction of the present-day geomagnetic field in Alcázar.

TABLE II

SUMMARY OF MEASUREMENTS: NUMERICAL DATA FROM THE SITES[1]

Age	*Site*		*Samples*	*Correction for the dip*	*a.c.*	*Mean direction*[2] *and a_{95} for a given site:*		
						D	*I*	a_{95}
Triassic	(I)	El Doncel	VAL 1–25	190– 7	750–3,000	6	22.5	3
			ALC 21–23	190– 7	750–3,000			
Triassic	(2)	Crra N 420	VAL 26–34	120– 4	750–3,000	353	23	6
			ALC 1– 3	120– 4	750–3,000			
Silurian	(IA)	Tordelloso	CAN 1– 7	212– 30	200–2,000	131.5	17	8
Silurian	(IB)	Tordelloso	CAN 8–13	257– 30	300–600	174	18	16
Silurian	(II)	Peña Negra	CAN 14–23	315– 45	300–1,000	161	11	5.5
			VAN 24–30	315– 45	300–2,000	163.5	15.5	5.5
Silurian	(III)	Tordelloso	VAN 3–10	280– 30	200–1,000	161	19	3.5
Silurian	(IV)	Tordelloso	VAN 11–19	280– 30	300–1,500	163	27	8
Silurian	(V)	Almadén	VAO 7–11	102– 70	150–200	121	20	11
Silurian	(VI)	Alma-denejos	VAO 17–21	79–120	150–500	140	25	19.5

[1] From each site some samples were not measured. They will be kept for experiments in the future. a_{95} = semiangle of the cone of 95 % confidence.
[2] Corrected for the present-day local magnetic declination.

The initial measurements on the N.R.M. of these samples revealed for both sampling sites a direction with an inclination much lower than the inclination of the present-day geomagnetic field (Fig.3A). Progressive demagnetization with a.c.magnetic fields up to 3,000 Oe (peak value) proceeded more or less the same for all samples. The results of these progressive demagnetizations are illustrated in Fig.4. During treatment with a.c. magnetic fields with strengthes up to values varying between 750 and 1,500 Oe a component is eliminated which conforms to the present-day geomagnetic field and thus is secondary. Between 1,500 and 3,000 Oe the demagnetization reveals another component with a direction similar for all samples; the latter is the characteristic component (plotted in stereographic projection in Fig.3C).

In most cases the siltstones had not completely lost their remanent magnetization after treatment with a.c. magnetic fields up to 3,000 Oe and the decrease of the magnetic vector did not always take place in a direction that passed through the origin: a small very hard magnetization may still be present in most samples. The fact, however, that the end point of the magnetic vector ran for all samples along a straight line which passed the centre of the coordinate system at both sides or did not lead exactly through the origin indicates that a deviation caused by the induced magnetism and due to the anisotropy in the magnetization or

sample shape can be held responsible, as small very hard magnetizations are not likely to have directions at random in a group of samples from one site (ZIJDERVELD, 1967a).

RESULTS OF THE SILURIAN SAMPLES

The Atienza group (samples VAN 1–34, CAN 1–23)

The intensity of these andesites varied between 60.10^{-6} and $7{,}700.10^{-6}$ e.m.u./cm^3. They have a remarkable high Q value, the ratio between the N.R.M.

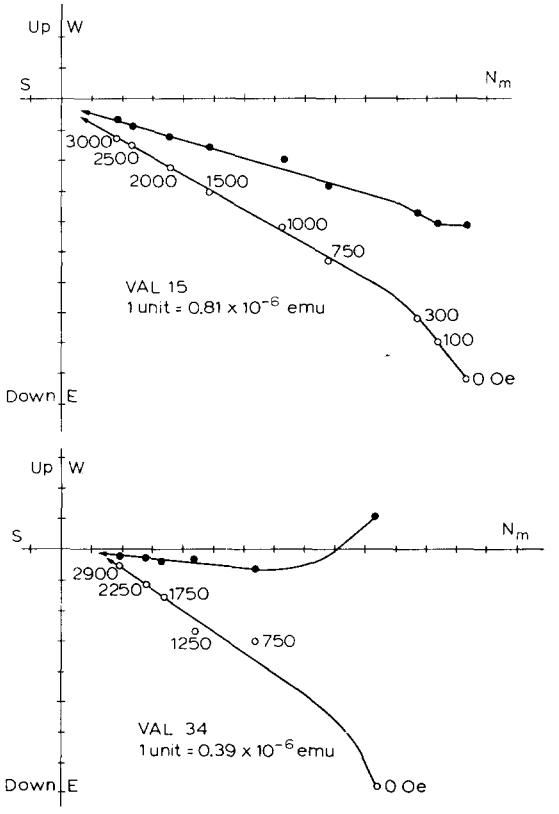
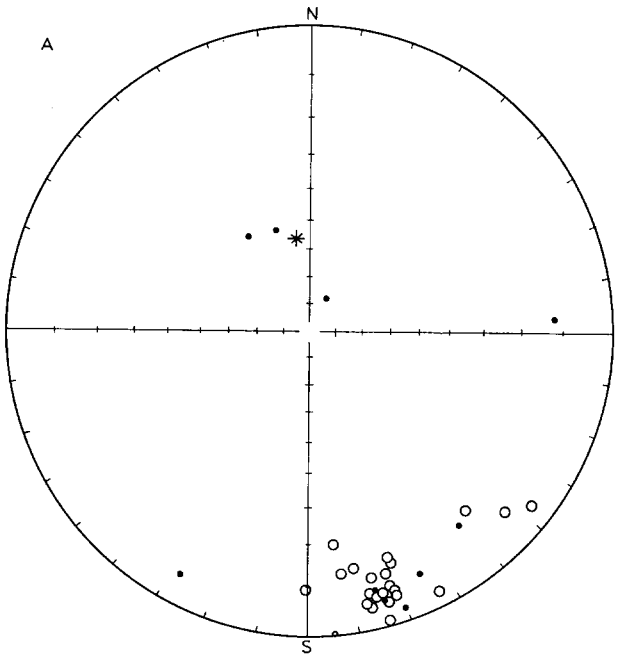
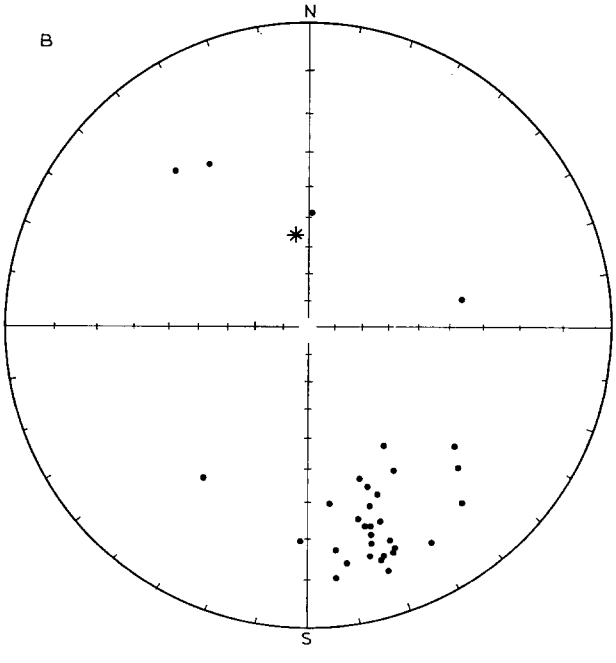


Fig.4. Demagnetization of two Triassic siltstone samples. Plotted points represent successive positions—in orthogonal projection—of the end of the magnetic vector during progressive demagnetization.

● = projection on the horizontal plane; ○ = projections on the north–south vertical plane. Not corrected for the present-day geomagnetic declination (July, 1966: 7.5°W). N_m = magnetic north.


A
N
S


B
N
S

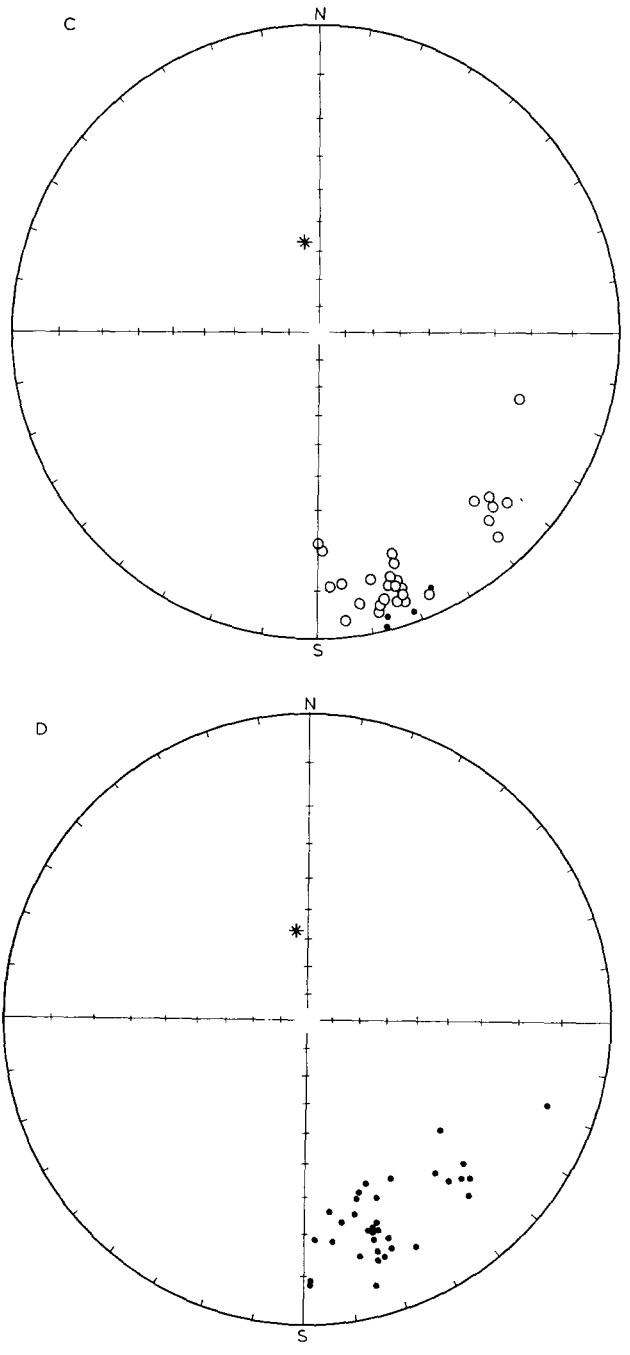


Fig.5. Silurian andesites from Atienza. Stereograms showing the directions of magnetic vectors: A. N.R.M., without tectonic correction; B. N.R.M., after tectonic correction; C. The characteristic components, eliminated between 300 and 1,000 Oe, without tectonic correction; D. The characteristic components, after tectonic correction.

● = north-seeking poles pointing down; ○ = north-seeking poles pointing up; ★ = the direction of the present-day geomagnetic field in Atienza.

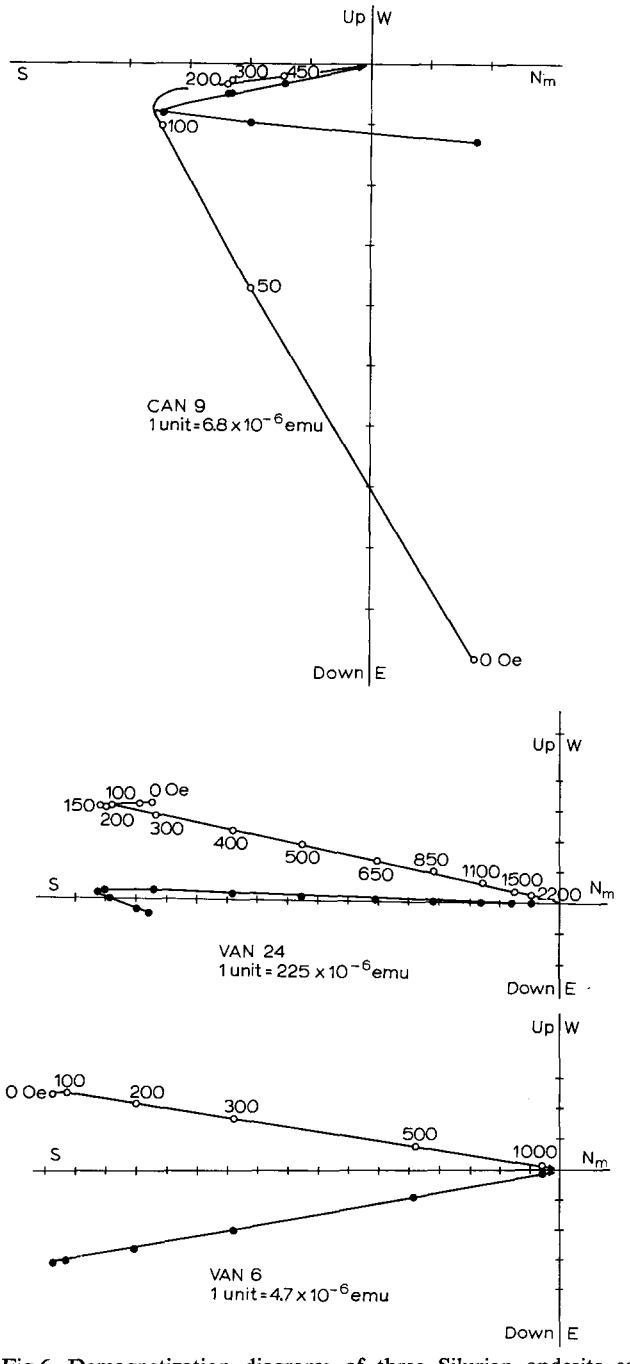


Fig.6. Demagnetization diagrams of three Silurian andesite samples from Atienza. Plotted points represent successive positions—in orthogonal projection—of the end of the magnetic vector during progressive demagnetization.

● = projections on the horizontal plane; ○ = projections on the north–south vertical plane.

Not corrected for the present-day geomagnetic declination. (July, 1966: 7.5°W). N_m = magnetic north.

and the induced magnetization $k.H$ (k = susceptibility, H = the strength of the ambient field[1]), see KOENIGSBERGER (1938).

The directions of the N.R.M.s are plotted in Fig.5A. The process of demagnetization did not shift the magnetic vectors very much, with the exception of the samples of one of the sites near Tordelloso (site *I A*). The secondary component of all these andesites is eliminated in a.c. magnetic fields below 200 Oe; ten samples have been progressively demagnetized (see Fig.7) and the remaining samples have been demagnetized in only four steps (300, 400, 500 and 600 Oe). The decrease of the magnetic vector then passed through the origin, revealing that only one magnetization is left, wihch appeared to be characteristic for these Silurian rocks. The directions obtained in demagnetization are plotted in Fig.5C, while in Fig.5D the correction for the geological dip has been applied.

The average direction has been computed giving unit weight to the means for each site (six sites, D: 159°, I: 18.5° and α_{95}: 12°). The semiangle of the cone of 95% confidence giving unit weight to samples ($N = 33$) is 5°.

The Almadén group (samples VAO 7–11, VAO 17–21)

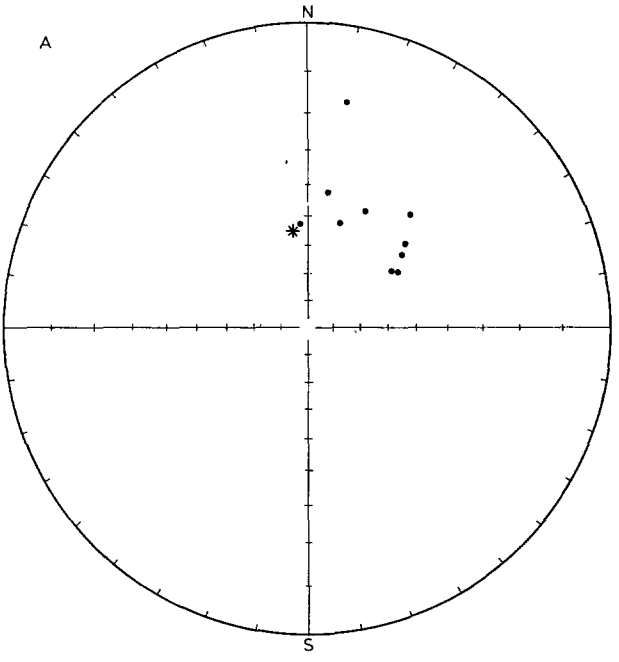
The intensity of the N.R.M. of the five samples from Almadén varied between 430.10^{-6} and 590.10^{-6} e.m.u./cm^3, and of the five samples from Almadenejos between $1{\cdot}1.10^{-6}$ and $30{\cdot}10^{-6}$ e.m.u./cm^3.

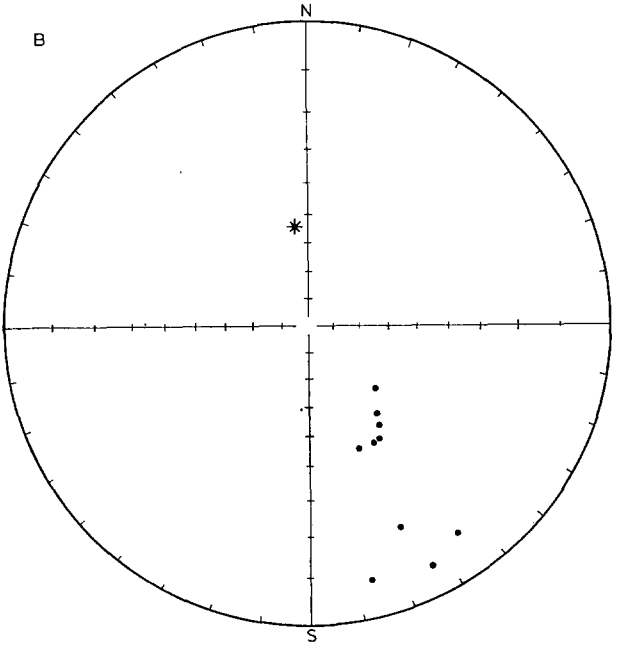
As shown in Fig.7A, the directions of the N.R.M.'s are rather scattered, but after a.c. magnetic field demagnetization the two groups of the two sites can be distinguished (Fig.7C); moreover the different correction for the geological dip of the strata brings the two groups together (Fig.7D). These ten samples, however, did not yield a fine cluster like that of the Atienza group ($\alpha_{95} = 11°$, giving unit weight to the samples, $N = 10$).

The samples from the site near Almadén gave a remarkable result in the process of demagnetization. As shown in Fig.8A and 8B, at least three directions of magnetization can be recognized (between 0 and 50 Oe, between 150 and 250 Oe and between 450 Oe and the origin). The first and the latter have more or less the same direction and conform to the present-day geomagnetic field. One is apt to call the second component the characteristic one for these five samples, and it is this component which has been plotted in the stereographic projection of Fig.6D. (See for an extensive description of the phenomenon of two secondary magnetizations ZIJDERVELD, 1967a, p.283 and fig.15, p.285.) The second (characteristic) component is very well consistent with the characteristic component of the samples VAO 17–21 from Almadenejos, after the correction for the geological dip.

From the same region six samples from Devonian sediments and eight more samples from Silurian volcanics and sediments were collected, but the intensities of their N.R.M.'s were too low to be properly measured and demagnetized.

[1] In the case of our measurements 0.44 Oe.


A
N
S


B
N
S

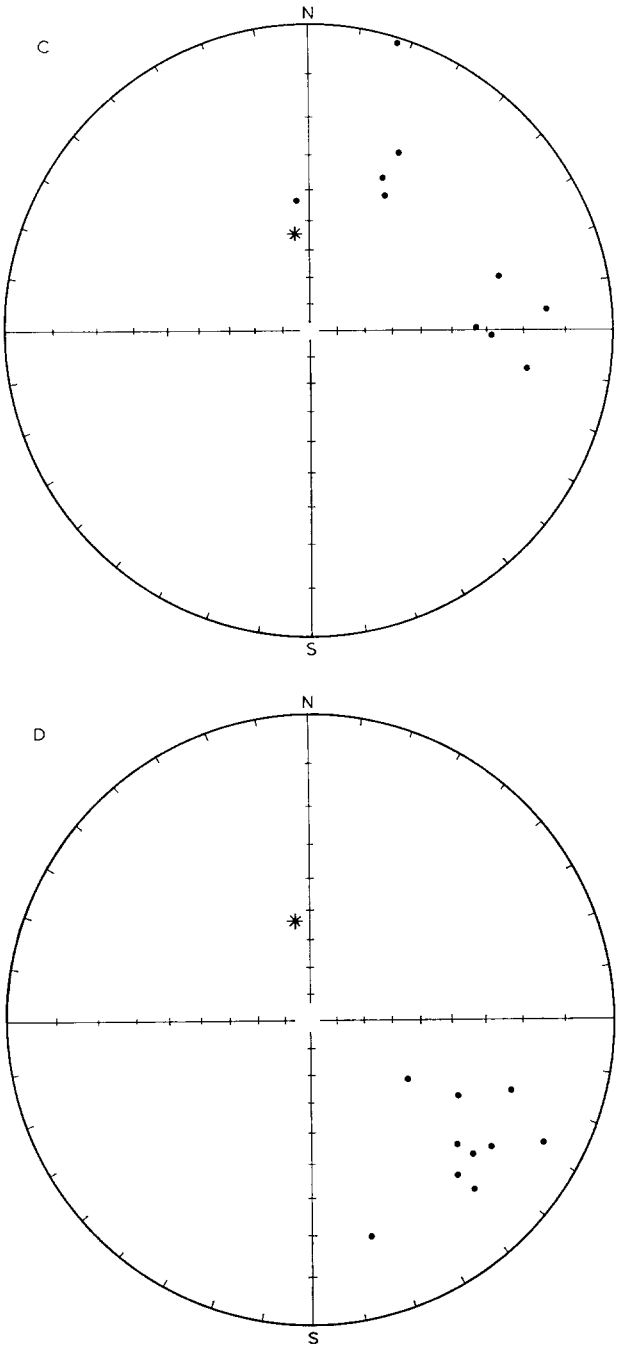


Fig.7. Silurian volcanic rocks from Almadén and Almadenejos. Stereograms showing the directions of magnetic vectors: A. N.R.M., without tectonic correction; B. N.R.M., after tectonic correction; C. The characteristic components, without tectonic correction; D. The characteristic components, after tectonic correction.

● = north-seeking poles pointing down; ★ = the direction of the present-day geomagnetic field in Almadén.

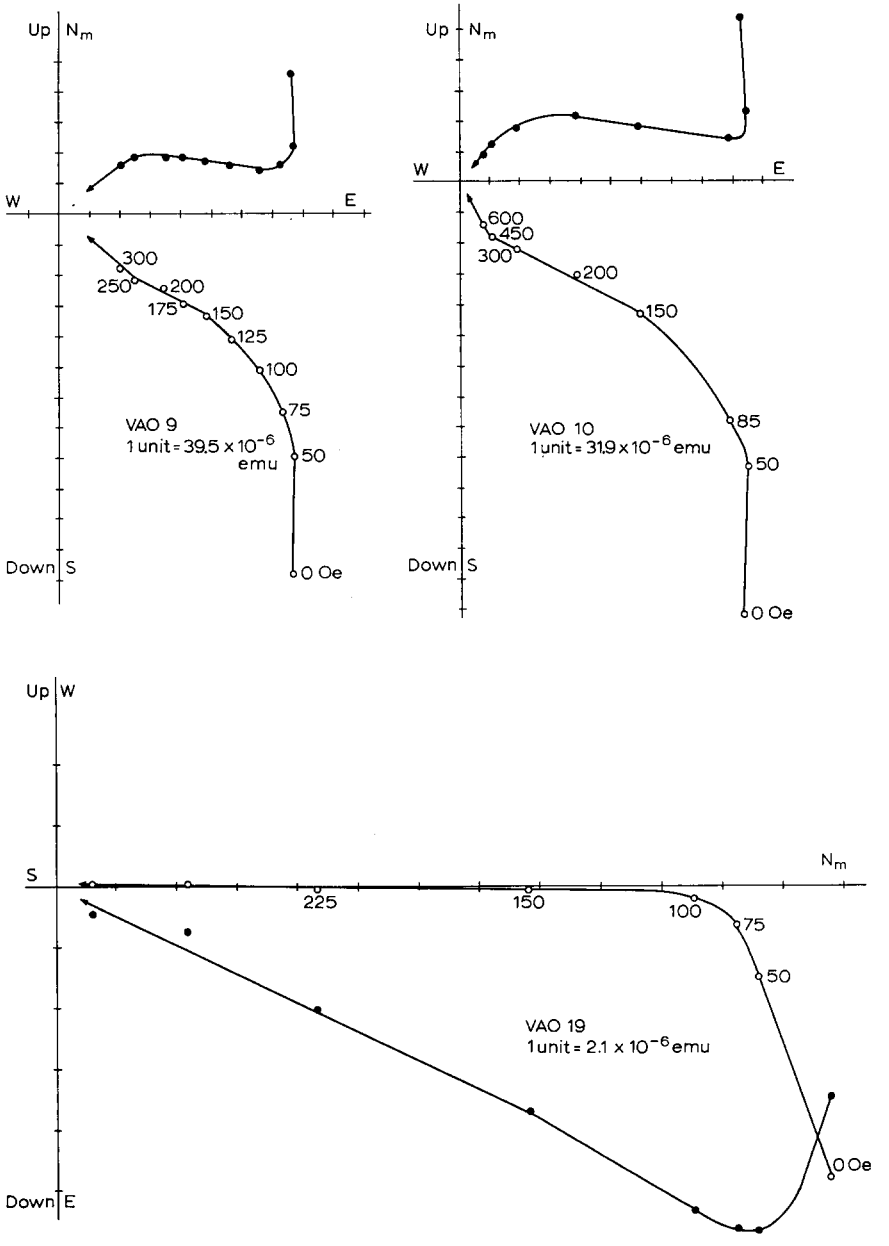


Fig.8. Demagnetization diagrams of three Silurian volcanic rocks, from Almadén (VAO 9 and 10) and Almadenejos (VAO 19). Plotted points represent—in orthogonal projection—successive positions of the end of the magnetic vector during progressive demagnetization. Full symbols represent projection on the horizontal plane, open symbols projections on the east–west (VAO 9 and 10) or north–south (VAO 19) vertical plane.

Not corrected for the present-day geomagnetic declination (July, 1966: 8°W). N_m = magnetic north.

CONCLUSIONS

In Table III the data are summarized; the Silurian directions are regarded as reversed.

Since more palaeomagnetic investigations in central Spain and Portugal are still in progress, these results will finally be discussed in more detail. With the data now available, however, we may conclude that the palaeomagnetic directions obtained from Spain yield a rather marked disagreement between these directions and the coeval palaeomagnetic results from northern and central Europe (stable or extra-Alpine Europe), viz. that a difference in declination is apparent.

In Table III the palaeomagnetic pole positions from stable Europe for Silurian and Triassic are summarized; in Fig.9A the directions computed from these pole positions for Madrid (4°W 40.5°N) can be compared with the data obtained from Spain.

Considering these data we see that the divergence between the results from stable Europe and the Spanish results mainly exists in the declination; the simplest conclusion that can be drawn from this behaviour is to make a counter-clockwise rotation of Spain responsible for the deviation, see also GIRDLER (1965).

Fig.10 depicts a hypothesis of DU TOIT (1937), CAREY (1958) and BULLARD et al. (1965) concerning a counter-clockwise rotation of 32° of Spain, based on the best fit in an ancient continental configuration and on other reasons. Shifting Spain in this position brings Madrid at 5°W 42.5°N in the present geographic coordinates. In Fig.9B the directions of stable Europe from Fig.9A are tentatively recomputed for this ancient hypothetical position of Madrid. A good agreement both for Triassic and Silurian palaeomagnetic directions appears after rotating Spain 32°. Particularly the Swedish Silurian results (MULDER, 1965) are in good harmony with the Spanish Silurian.

As up until now only few reliable palaeomagnetic directions from rocks, which have been cleaned with thermal or a.c. magnetic field demagnetization, are available from stable Europe, it is still impossible to determine the accurate amount of the rotation. But, as shown in Fig.9A and 9B, the paleomagnetic data from Spain support clearly the topographically predicted rotation.

ACKNOWLEDGEMENTS

I wish to express my appreciation for the continuous interest of Professor Dr. M. G. Rutten who stimulated this investigation.

Many thanks are due to Professor Dr. J. Veldkamp, under whose supervision this work has been completed, and to Dr. J. D. A. Zijderveld for their critical remarks.

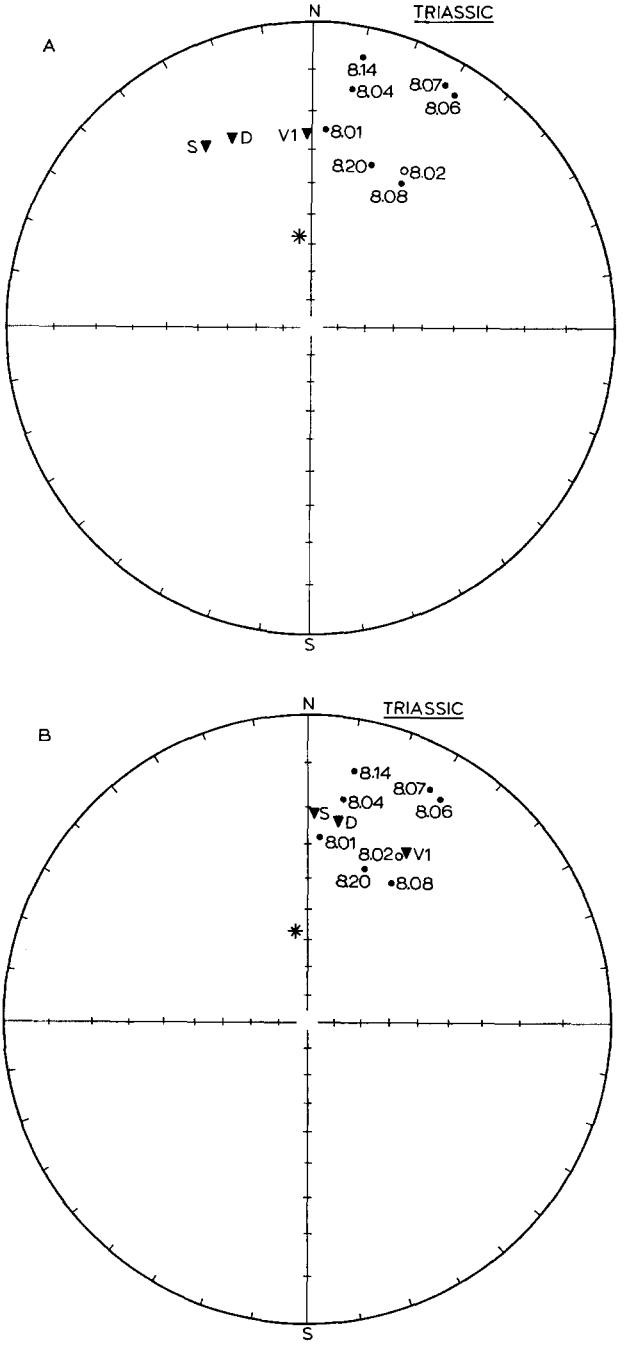


Fig.9. Stereographic projection of Triassic and Silurian directions, as found in paleomagnetic research in stable Europe (full or open circles) and Spain (triangles). The directions are computed for: A. The present-day position of Madrid (4°W 40.5°N); B. A possible ancient position of Madrid (5°W 40.5°N), together with a rotation of Spain of 32°, as depicted in Fig.10.

The numbers refer to IRVING (1964, *4.01–4.07* and *8.01–8.20*) and to MULDER, 1965 *ZMA–ZMF*). See also Table III.

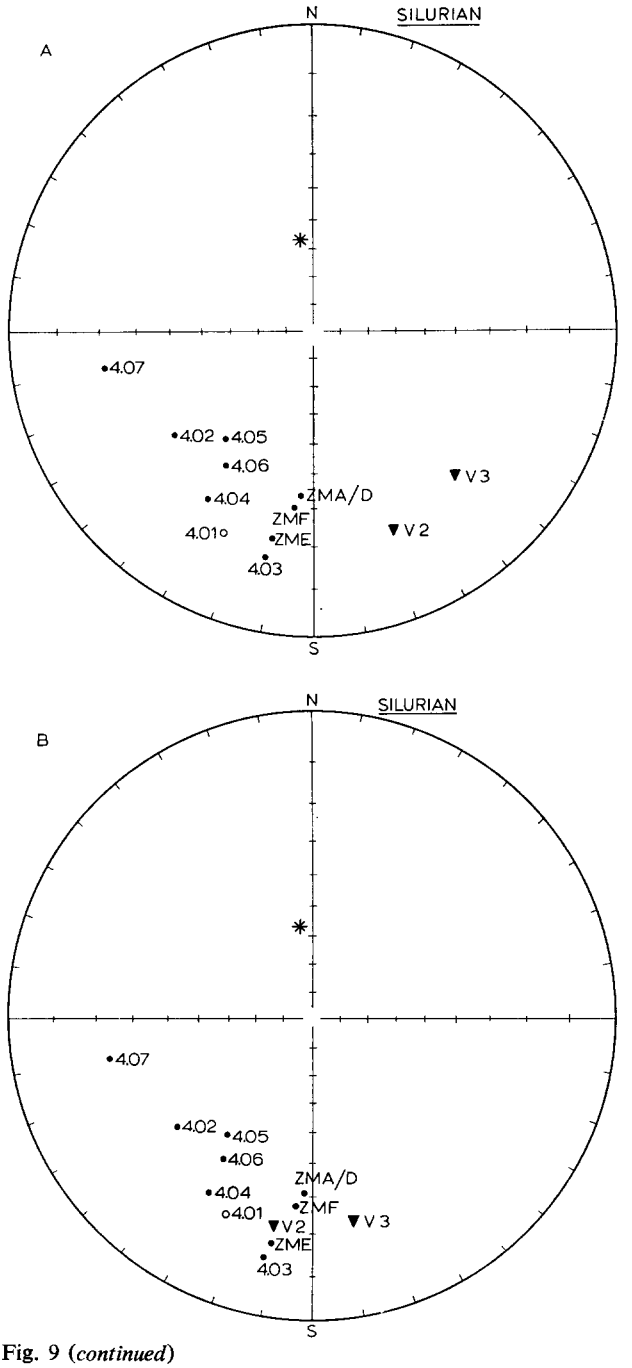


Fig. 9 (*continued*)

For the Spanish results (triangles) we refer to: VAN DONGEN (1967): Lower Triassic from the eastern Pyrenees, indicated as *D*; SCHWARZ (1962, 1963): Permo-Triassic from the central Pyrenees, indicated as *S*; Present paper: Triassic from central Spain, indicated as *V 1*; Silurian from central Spain (Atienza), indicated as *V 2*; Silurian from central Spain (Almadén), indicated as *V 3*.

TABLE III

MEAN DIRECTIONS OF MAGNETIZATION AND DEDUCED ANCIENT POLE POSITIONS

	Rock type	N_1	N_2	*D*	*I*	(α_{95}) *samples*	(α_{95}) *sites*	*Ancient pole position*	
Lower Triassic—Alcázar	S	2	39	359.5	23.0	6		63°N	177.5°E
Upper Silurian—Atienza	B	6	33	159	18.5	5	12	35.5°N	157°W
Upper Silurian—Almadén	B	2	10	130.5	22.5	11		21°N	132°W

OTHER DIRECTIONS FROM SPAIN AND ANCIENT POLE POSITIONS

	Rock type	N_1	N_2	α_{95}	*Demagnetization*	*Ancient pole position*	
Lower Triassic (VAN DONGEN, 1967)—Seo de Urgel, eastern Pyrenees	S	1	4	11	a.c. magnetic field	54.5°N	142°W
Permo-Triassic (SCHWARZ, 1962, 1963)—Oza, central Pyrenees	B	3	14	5.5	a.c. magnetic field	50.5°N	133°W

FOR COMPARISON SOME TRIASSIC AND SILURIAN DATA FROM STABLE EUROPE. (THE NUMBERS REFER TO IRVING, 1964.)

8.01	France-Vosges, Low. Triassic	S	3	9	23		62°N	167°E
8.02	France-Vosges, Low. Triassic	S	7	61?	11		28°N	143°E
8.04	Germany, Low. Triassic	S	5	25	13		55°N	159°E
8.06	England, Upp. Triassic	S	9	43	12	a.c. + thermal	43°N	131°E
8.07	England, Upp. Triassic	S		35		a.c. magnetic field	44°N	134°E
8.08	Scotland, Triassic	S		41?	21		54°N	118°E
8.14	U.S.S.R., Low. Triassic combined	S					51°N	159°E
8.20	U.S.S.R., Triassic Siberian traps comb.	B					60°N	133°E
4.01	Wales, Ludlow, Silurian	S		7			52°N	134°E
4.02	U.S.S.R., Dniester, Silurian			12			14°N	124°E
4.03	U.S.S.R., Dniester, Silurian						40°N	160°E
4.04	U.S.S.R., 4.02 and 4.03 combined						29°N	140°E

TABLE III (*continued*)

			Rock type	N_1	N_2	α_{95}	*Demagnetization*	*Ancient pole position*	
4.05	U.S.S.R., Ural, Silurian		B		6			16°N	140°E
4.06	U.S.S.R., North Urals, Silurian					21		22°N	141°E
4.07	U.S.S.R., Lena, Silurian		S		12			0°	101°E
ZMA/D	Sweden, Vänern Silurian	MULDER (1965)	B	4	24	8.5	a.c. magnetic field	33°N	169°E
ZME	Sweden, Vänern Silurian	MULDER (1965)	B		6	3	a.c. magnetic field	38°N	162°E
ZMF	Sweden, Vänern Silurian	MULDER (1965)	S		6	3	a.c. magnetic field	31°N	172°E

N_1 = number of sites, N_2 = number of samples; D and I = declination and inclination in degrees; α_{95} = semiangle of the cone of 95% confidence; *Rock type*, B = basic igneous rocks, S = sediments.

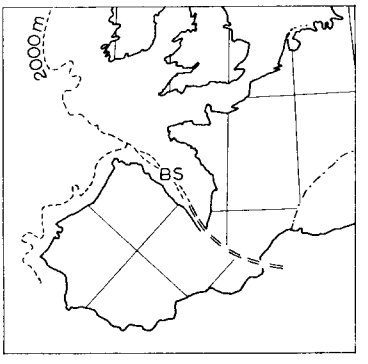


Fig.10. The minimum rotation and translation of Spain, as indicated by the best fit (CAREY, 1958; IRVING, 1964, p.255 and BULLARD et al., 1965). The Biscay sphenochasm (BS) is closed at the 2,000 m isobath.

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